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is zorgzaam en
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FLEMISH HEAT ACTION PLAN

A health plan about heat



Update 2024

DEPARTEMENT
ZORG

www.departementzorg.be

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1 PROBLEM

Heat is the deadliest natural phenomenon in Europe. In Flanders too, the greatest impact of climate change on health and well-being is expected to be heat-related. Those who are vulnerable feel these effects even harder. Such as the elderly, children and the sick.

Health Effects:

- greater risk of (deterioration of) kidney disease, respiratory disease, and cardiovascular disease such as strokes and drops in blood pressure;
- less sleep and therefore more fatigue, less alertness and more severe psychological distress issues;
- more premature deaths, for example due to the combination of heat and heart disease; • more cases of overheating, dehydration and a lack of salt in the blood; • increased risk of premature birth and increased blood pressure in pregnant women; • enhanced effects of heat with, for example, diuretic medications and some medicines for psychiatric disorders;
- greater risk of allergies, cancer and hormone disruption due to higher concentrations of pollutants substances in the indoor air.

2 FRAME

2.1 NATIONAL OZONE - A FINDING PLAN

Since 2005, Belgium has had [a national ozone and heat plan](#). This plan describes how the Belgian population will be warned when public health problems are expected due to periods of very hot weather (e.g. heat wave). This plan also takes ozone concentrations into account because they have an additional negative effect on public health during these warm periods. The combination of both temperature and ozone criteria makes the Belgian plan unique.

This plan is followed up within the National Environmental Health Action Plan (NEHAP). Representatives of all services involved in Belgium (Federal, Region and Community) monitor the implementation and evaluation of the national plan. Until 2014, the Federal government was responsible for the implementation of this plan. More specifically, they were responsible for sending out warnings and raising awareness among the population and the actors involved. The regional authorities provided communication support.

The 6th state reform regionalized the powers for the implementation of the National Ozone and Heat Plan (figure 1). From now on, for Flanders, it is the Department of Care that is responsible for sending out heat warnings and raising awareness of both the general population and organizations and professionals that work with vulnerable groups. The implementation of this is described in the Flemish Heat Action Plan.

However, the alarm phase, which is regarded as a crisis phase, remains a Federal competence. The regional authorities do support communication during this phase.

Since 2015, the national ozone and heat plan has bundled all regional and federal initiatives on ozone and heat. Coordination between all authorities continues to take place within the NEHAP.



Figure 1: Overview of the National Ozone and Heat Plan



2.2 FLEMISH HEAT ACTION PLAN

This plan outlines how and when vulnerable target groups - and professionals who work with those target groups - should be informed and warned before and during heat periods in Flanders. The action plan provides tools for professionals to take preventive actions to prevent health effects due to heat. The Flemish Heat Action Plan consists of a vigilance and warning phase.

2.2.1 Vigilance phase

From May 15 to September 30, the maximum temperature forecasts in Uccle are monitored by the Royal Meteorological Institute (RMI). The Interregional Cell for the Environment (IRCEL) monitors ozone concentrations and ozone forecasts during the same period. IRCEL bundles this data and tests it against the criteria of the warning phase and alarm phase. When the criteria are met, they alert the authorities involved.

During the vigilance phase, the Department of Care asks professionals who work with vulnerable target groups to prepare for a possible period of prolonged warm weather. On May 15, the Department of Care will send a newsletter to these professionals, drawing attention again to the problem and referring to the campaign website www.warmedagen.be.

The necessary supporting information is provided on this website. In order to build a sustainable policy around the heat problem, which involves thinking beyond ad hoc measures during a heat period, the Department of Care, with the support of the Medical Environmental Scientists of the Logos, encourages local authorities and professionals who work with vulnerable groups to draw up a local health plan for Warm Days. The preparation of a local health plan for Warm Days includes the elaboration of momentary actions (communication of recommendations, providing extra care, etc.) on the one hand and long-term intentions (spatial planning, greening of the environment, etc.) on the other that make a sustainable policy possible. To this end, coordination, networking and information exchange between involved parties within an organization and between

2.2.2 Warning phase The

warning phase is declared when possible health effects due to very warm temperatures are expected. The general population and professionals who work with vulnerable target groups are then asked to take extra measures (which are ideally included in a local Health Plan Warm Days) to protect themselves and risk groups.

The Department of Care sends out a newsletter for this purpose and will inform the general population through the press.

Since 2017, the criteria for starting the warning phase have been simplified. For example, the 5-day forecast of the maximum temperature in Uccle is examined. The ozone concentrations are no longer included as a criterion for this phase. The advantage of this change lies in the fact that organizations are no longer warned on the first very hot day, but one or two days in advance. The organizations and professionals that work with vulnerable groups therefore have more time than before to prepare themselves organizationally for additional measures.

Warning phase criterion: The predicted maximum temperatures in Uccle are followed. The warning phase is declared when Tcumul, on Day 0, is greater than or equal to 17°C. Tcumul on Day 0 is the sum of the difference between the 'maximum forecast temperature' and 25°C for the next five days (Day +1 to Day +5), taking only the positive differences into account. Example:

Day	Predicted maximum temperature (°C) 25	Number of °C above 25°C	Tcumul (°C) (day 1 to day 5) 22
Day 0 (today)			
Day 1	27	2	
Day 2	29	4	
Day 3	31	6	
Day 4	30	5	
Day 5	30	5	

The warning phase starts on day 0. The communication will also state when the effective heat period will start. This heat period starts from the first day, in the warning phase, where 28°C is forecast. In the example above, the heat period will start on day 2.

2.3 FEDERAL ALERT PHASE

Since the creation of the national ozone and heat plan, the alarm phase has only been declared once, namely in the summer of 2020. This phase is considered a crisis phase and is important to be able to take additional measures on top of the measures already taken during the warning phase.

When all criteria of the alarm phase are met, it will not be started automatically. Experts and policymakers will consult with each other to consider whether an upscaling to the highest phase of the national ozone and heat plan is necessary.

Since 2019, the criteria for the alarm phase have been reviewed. This makes it easier to meet the criteria, but the consideration of whether the alarm phase is actually declared by experts and policymakers remains.

Alarm phase criteria:

- 1. The temperature criterion for the warning phase has been reached.**

IN

- 2. The maximum predicted temperature for the day itself is greater than or equal to 28°C.**

IN

- 3. An hourly average was recorded the previous day at at least one ozone measuring site in Belgium
ozone concentration higher than 180 $\mu\text{g}/\text{m}^3$ (EU information threshold) is measured AND an hourly average
ozone concentration higher than 180 $\mu\text{g}/\text{m}^3$ (EU information threshold) is predicted for the day itself in a
significant part of the country.**

IN

- 4. The measures already taken must be strengthened. When the alarm phase is started, communication will be done
by the Federal government. The regional authorities (Department of Care for Flanders) will support
communication with organizations and professionals.**

3 FLEMISH HEAT ACTION PLAN

3.1 PURPOSE OF THE FLEMISH HEAT ACTION PLAN

The Flemish Heat Action Plan recognizes the urgency to protect vulnerable groups against the health risks caused by heat waves and high temperatures. This plan aims to implement measures that reduce the impact of heat on public health and strengthen the resilience of communities.

Goals:

- Protect vulnerable groups, including the elderly, young children, people with chronic diseases and socio-economically vulnerable individuals, from heat-related health problems.
- Developing an integrated approach for monitoring heat periods, risk communication and intervention, aimed at preventing heat-related deaths and health complications. • Carrying out awareness campaigns on the health risks of heat and the available preventive measures, aimed at the general public and specific target groups.
- Disseminate educational materials and guidelines for heat-resistant behavior, such as hydration, avoiding excessive sun exposure and seeking cool places.

Currently, the emphasis is still on actions just before and during warm periods. However, it is not only important to take action during these periods, but also to work proactively. To protect buildings and the environment against heat stress, measures are needed that work in the long term. Our environment must be adapted to limit heat stress and preferably avoid it as much as possible. We then no longer speak of a heat action plan, but a heat adaptation plan. An additional objective can be included for this:

- Creating resilient and heat-resistant communities through urban planning, green infrastructure and access to cool spaces during heat periods.

3.2 ONGOING ACTIONS

3.2.1 Awareness campaign 'Hot days, take care'

An awareness campaign about heat and health has been running since 2018. This campaign consists of several parts:

- An informative website for citizens and for organizations or professionals who work with vulnerable groups •

Awareness materials tailored to different target groups • Support materials tailored to different target groups • Tools that visualize the heat problem

The campaign emphasizes the visual aspect to make the messages as accessible as possible for everyone.



The campaign currently focuses on the following vulnerable target groups and/or the organizations or professionals that work towards these target groups:

- The elderly with a focus on elderly care facilities •
- Young children with a focus on childcare initiatives and schools (primary and secondary) • Local authorities: they have a dual role as operator of facilities with vulnerable groups and as a general sensitizer •
- Health and welfare services that can come into contact with socially isolated people • General practitioners as contact persons for patients (due to underlying conditions or the taking certain medications) •
- Home care services and informal caregivers • Event organizers (cultural and sports) •
- Athletes and sports organizers (sports clubs)

Heat within the working environment is not included as a vulnerable context because it is included under labor regulations (FOD WASO).

3.2.1.1 Website

Interested citizens or professionals will find the necessary information about heat and health on the website www.warmedagen.be. The website forms the basis for the awareness campaign, but also for the Flemish Heat Action Plan. 'Warm days, taking care' is profiled as a brand to guarantee recognisability and reliability of the content.



3.2.1.2 Awareness materials A range of

campaign materials can be found on the [website](#). All materials are downloadable and some materials can be ordered. They form the backbone for the recognizability of the theme. At the same time, emphasis is placed on knowledge transfer to stimulate behavioral change. Accessibility and clarity should increase the reach of vulnerable groups. These materials are generally useful for all target groups.

Some are tailor-made for a specific target group.

Sensitization materials (2024)	
Can be ordered in printed form and downloadable	
Brochure 'Hot days, taking care'	General brochure. Can be transformed into a windmill.
Poster 'Lots of sun? Also think of grandma Yvonne.'	
'A lot of sun? Also think of Grandpa	
Leon.' Poster 'High temperatures? Think about your	
neighbors.' Informative poster in general	
Informative poster 'target group children'	
Informative poster 'target group elderly'	
Informative poster 'target group of athletes'	
Contact and medication sheet	Sheet with tips and place to record important contacts (e.g. GP) and medication used.
Drinks menu	Postcard format and tear-off block (with 50 drink cards).
Downloadable	
Poster 'urine meter'	Fun poster to draw attention to the color of urine as a measure of sufficient drinking.
General flyer	
Flyer 'socially isolated'	Small size of the contact and medication sheet.
Drinks menu – different sizes	
Digital screens – Clearchannel City Play	Digital images for local authorities. To be used on public land.
Digital screens – pharmacists, general practitioners	Digital images for the digital screens at pharmacies or waiting rooms, among others. general practices. 21
Images for social media	images that can be used on social media platforms.
Advertorials	Standard texts about with the main messages of the campaign. Can be used for information sheets.
Pictures	13 derivative images of the campaign (banners) 31 icons
Icons	from the posters that can be used separately.

3.2.1.3 Support materials In addition to the

campaign materials that raise awareness and promote behavioral change, there are also a number of [support materials](#) available. These materials provide more background on the issue of heat on health and are target group specific. For example, there are guidelines for drawing up a Hot Days health plan, checklist, inspiration documents and overview documents. The purpose of these materials is to help organizations and professionals who work with vulnerable target groups

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support in developing a Warm Days health plan and in formulating actions and objectives.

Support Materials (2024)	
Target group of local authorities	
Health plan guidelines	Manual for drawing up a health plan 'Warm Days'.
Inspiration matrix	List of possible actions to prevent health effects of very hot days.
Guide to local government events	Clarification of the role of local authorities in the organization of events within the municipality.
group of children.	
Health plan guidelines	Manual for drawing up a health plan 'Warm Days'.
Health risks of heat in children	Overview document with vulnerabilities and symptoms of heat stress in children List of
Inspiration matrix for schools	possible actions for schools to prevent health effects of very hot days.
Inspiration matrix for childcare initiatives	List of possible actions for childcare initiatives to prevent health effects of very hot days.
Problem analysis and intervention options	Overview of the main messages and recommendations of the campaign.
Checklist of initial situation of schools	This checklist can be used to map out what the school is currently doing to prepare for hot days and protect students against heat.
Checklist initial situation of childcare initiatives	This checklist can be used to map out what a childcare initiative is currently doing to prepare for hot days and protect children from heat.
Target group elderly	
Health plan guidelines	Manual for drawing up a health plan 'Warm Days'.
Inspirational document	Overview of possible actions for elderly care facilities.
Risk factors for heat in the elderly	Vulnerability overview document.
Motivational interviewing	Tips and points of interest to promote behavioral change through motivational interviewing.
Target group general practitioners	
Guidance	Document with attention to risk factors in patients and undesirable effects of medication during heat periods.
Target group of socially isolated people	
Health plan guidelines	A step-by-step plan for organizations to reach people in social isolation on very hot days.
Inspirational document	Possible actions and points of interest to reach socially isolated people.



Vulnerability problem analysis	Overview of why the target group is vulnerable to heat.
Working with volunteers	Overview document with points to consider when working with volunteers. 3 inspiring
Practical stories	practical examples.
Target group home care services and informal caregivers	
Guidance	Manual for home caregivers and informal caregivers.
Digital information session for home care services	'webinar' for home care services.
Presentation for home care services	Presentation that can be used to inform home care services.
Flyer for home care workers	Points of attention regarding heat and vulnerable groups within home care.
Target group athletes and sports organizers	
Health advice for athletes	Document 'How can you protect yourself against the heat?'.
Health effects in athletes	Overview document of heat-related conditions, symptoms and risk factors.
Health plan guidelines for sporting events	Guidance and inspiration guide for sporting events.
Sporting events checklist	Scoring system with recommendations regarding adjustments or cancellations of sporting events.
FAQ list for sports services, sports clubs and sports camps	Overview of points of interest and possible actions based on a few questions.
Infographic sport activity	Checklist for sports activities.
Target group of events	
Guide to health plan for cultural events A step-by-step	plan for organizers of cultural events during the summer.
Risk checklist	Checklist 'What are the risks at my event?'
Possible health problems	Overview document 'Causes of health risks at events' Inspiration list
Inspiration list	'how are you prepared for hot days?'
Health matrix 'cooling measures'	
Health matrix 'drink enough'	
Possible partners	Document maps which local actors you can or should involve in the preparation of your event on very hot days.

3.2.1.4 Tools

Two tools were developed specifically for local authorities. These tools can provide added value in developing a Hot Days health plan and in substantiating the necessity of these plans and the choices of actions.

3.2.1.4.1 Heat impact tool

The heat impact tool shows the impact of heat both numerically and visualized on a map, for Flanders, at municipal level or for a selected area, for both the current and a future scenario (average over the period 2040-2059).

The proposed figures in the tool can be used in local social policy. They provide an indication of the severity of the consequences of heat on mortality in a municipality now and in the future. The figures can also be used to supplement climate adaptation plans with health figures. Especially municipalities where the aging population is increasing rapidly will have to focus more on heat prevention:



- Heat warnings
- Heat action plans
- Focus on green-blue spaces

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www.klimaatengezondheid.be/hitte-impacttool

Hittegerelateerde sterfte

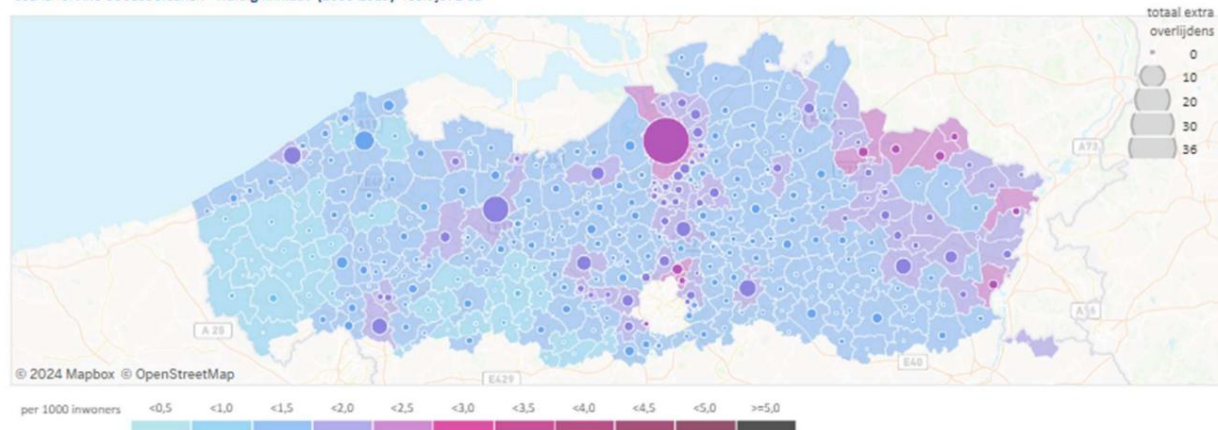
gebiedsindeling: selectie: toon op kaart:

doodsoorzaak: scenario: leeftijd:

klik links op kaartje op om kaartje te centraliseren na in-of uitzoom

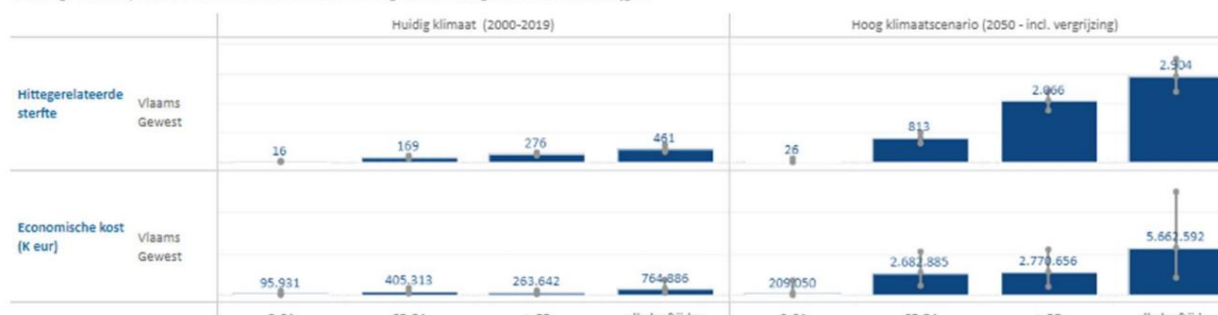
Extra overlijdens tijdens de zomer

scenario: Alle doodsoorzaken - Huidig klimaat (2000-2019) - leeftijd: ≥ 85



Overzicht huidig en toekomstig scenario voor alle leeftijden.

Klik één gemeente op de kaart aan om de data te filteren. Klik nogmaals om terug de Vlaamse data te krijgen.



3.2.1.4.2 Heat Vulnerability Map Not

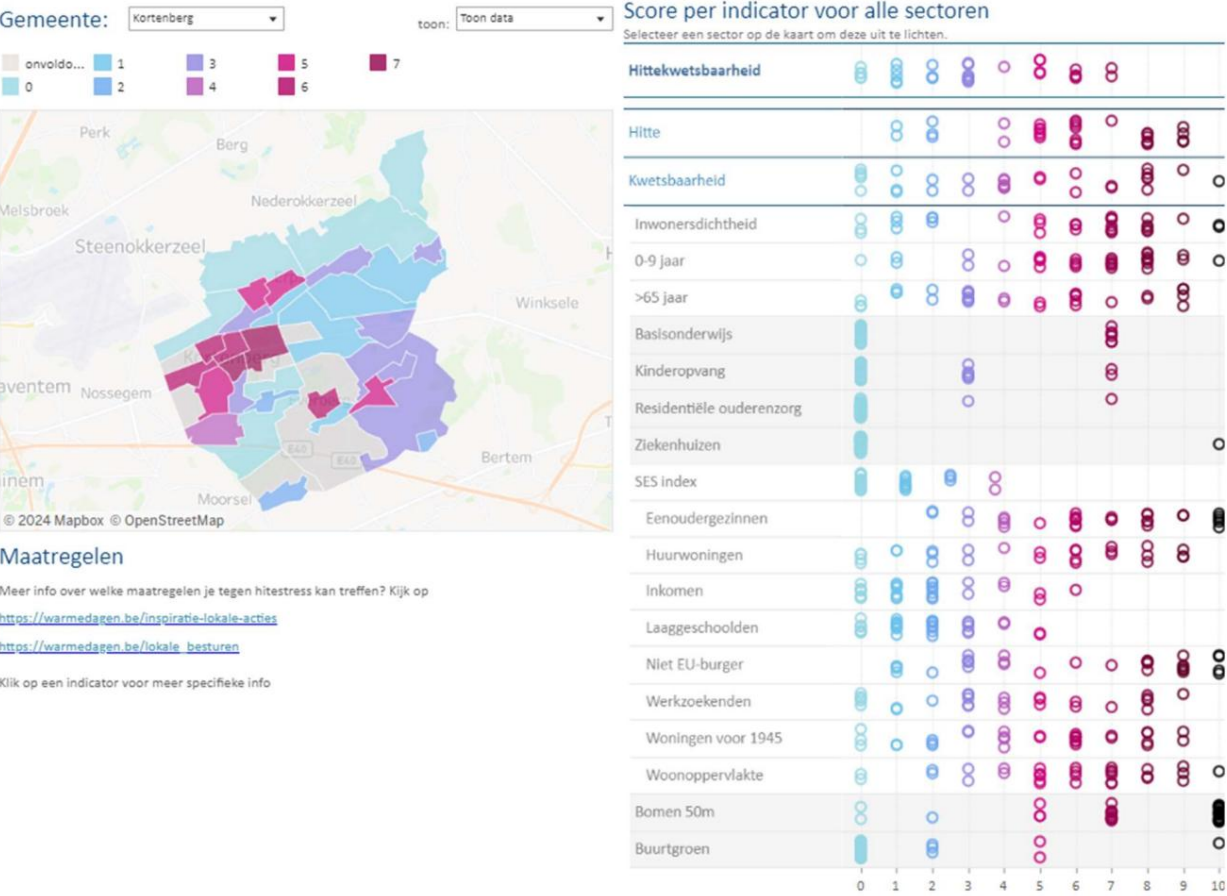
everyone experiences the same effects of heat. Some individuals or groups are more vulnerable to heat stress than others. The heat vulnerability tool maps these vulnerabilities using indicators. By combining the vulnerabilities with heat stress, you gain insight into those statistical sectors where heat vulnerability is greatest. In this way, the heat vulnerability map can serve as a guideline for (prioritizing) local policy on heat.

In other words, the heat vulnerability map shows where target groups with an increased vulnerability to heat stress live in Flanders, with specific attention to locations with an increased exposure to heat stress. And this goes down to the level of the statistical sector.

www.klimaatengezondheid.be/hittekwetsbaarheidskaart-lokaal

www.klimaatengezondheid.be/hittekwetsbaarheidskaart-vlaanderen

Hittekwetsbaarheidskaart van Kortenberg



3.2.1.5 Register for the Hot Days newsletter At the start

of the vigilance phase and at the start and end of the warning and alarm phase, the Care Department will inform all local authorities and organizations and professionals that work with vulnerable groups. These warning emails form the backbone of the warning system. The Department of Care collects e-mail addresses from its own databases and databases of other government agencies (such as the Growing Up Agency, Department of Education, etc.). Organizations and professionals can also register themselves via <https://zorg-en-gezondheid.be/nieuws/schrijven-u-in-op-de-nieuwsbrief-van-zorg-en-gezondheid>.

3.2.1.6 Support The

preparation of local health plans for Warm Days is not mandatory, with the exception of residential care centers, but is strongly encouraged. The [medical environmental scientists of the Local Health Consultation \(Logo\)](#) encourage local authorities to draw up a local health plan for Warm Days. They use their network function in health and environment to bring local actors together. The medical environmental scientists also provide substantive support in drawing up a local health plan for Warm Days. They do this primarily for local authorities, but other organizations that work with vulnerable groups can also make use of their substantive expertise. They also provide support in using the heat impact tool and the heat vulnerability map.

3.3 VIGILANCE PHASE (MAY 15 – SEPTEMBER 30)

During the vigilance phase, local authorities and organizations or professionals that work with vulnerable groups are asked to prepare for a possible warm period.

3.3.1 Newsletter starts vigilance phase

On May 15, a newsletter from the Department of Care will be sent to all local authorities and organizations or professionals that work with vulnerable groups. This newsletter once again draws attention to the problems and need for vigilance. Reference is made to the campaign website (www.warmedagen.be) and the awareness and support materials. Health problems during hot periods are brought to attention through an awareness message.

3.3.2 Continuous temperature monitoring

The Department of Care will closely monitor temperature developments every day from May 15. The Department of Care receives a daily update from the RMI in which the Tcumul (see above) is stated for a longer period. If the threshold value of 17°C is suspected to be exceeded, all internal preparations are made to switch to the warning phase. If the limit is actually exceeded, the Department of Care will effectively announce the warning phase.

Because the switch from the vigilance phase to the warning phase (or vice versa) can also take place during weekends, the on-call duty of the environmental health team is also involved. This way you can switch at any time if necessary.

3.3.3 Newsletter at the end of the vigilance phase

The vigilance phase will end on September 30. The Department of Care sends out a newsletter announcing this. All recipients are thanked for their efforts during the past summer season. Suggestions and comments on the newsletters, the campaign and campaign materials are solicited.

3.4 WARNING PHASE

As soon as the criteria of the warning phase are reached, the Department of Care receives an official message from the Interregional Cell for the Environment (IRCEL). Within the agreements of the national ozone and heat plan, they are authorized to confirm that the criteria have been achieved.

The Care Department takes care of the following actions:

3.4.1 Internal organization

Those responsible within the Department of Care, the office of the Flemish Minister of Welfare, Public Health and Family and the spokespersons of both are informed of the entry into the warning phase. During the weekend, the on-call team of the environmental health care team is called in to do the necessary. A script was drawn up for this.

3.4.2 Warning email

The Department of Care sends a warning email to all local authorities, organizations and professionals that work with vulnerable groups. This email asks you to prepare effectively for a warm period and to take actions to protect vulnerable people. The



main messages of the campaign 'Warm days, take care' are repeated. These key messages are: • Be prepared

- Drink enough
- Keep yourself cool
- How to keep your building cool • Take care of yourself and others

When the warning phase ends, all recipients will be informed.

3.4.3 Press release

The Department of Healthcare launches a press release to announce the start of the warning phase. Here too, the main messages are repeated and care for vulnerable people is highlighted.

3.5 ALARM PHASE

When the criteria of the alarm phase are met, IRCEL sends the necessary information by e-mail to the members of the Risk Assessment Group (RAG). The RAG is a group of experts, coordinated by [Sciensano](#), including experts from the [Department of Care](#), who evaluate the situation.

This takes into account the intensity, the possible duration of the heat periods, aggravating circumstances such as large-scale events or other problems and the measures already taken. Based on this evaluation, the RAG will provide advice on whether or not to scale up to the alarm phase and what additional measures are necessary. The RAG provides advice to the Risk Management Group (RMG), which consists of representatives from the cabinets of the competent ministers. Based on the advice of the RAG, the RMG activates the alarm phase or not and the necessary information is provided to the federal crisis center and the regional crisis centers. When scaling up, the press and the general population are also informed.

The Department of Care will communicate the decisions of the RMG, including what additional measures are needed, via a warning email to all local authorities, organizations and professionals that work with vulnerable groups.

During the alarm phase, crisis meetings are organized daily to closely monitor the situation. The Care Department will attend these crisis meetings to provide a picture of the situation in Flanders.

Within the NEHAP (National Environmental Health Action Plan), it was agreed that exceptional circumstances, for example a prolonged period of exceptionally high temperatures, but without high ozone concentrations, could lead to a decision to put the RAG to work and evaluate the situation. This is done in consultation with all parties involved in the national ozone and heat plan.

